

# Cisco Packet Tracer Project 1

## Design and Implementation of a Small Network for Accounts and Delivery Departments

### Objective

The objective of this project is to design a small office network using Cisco Packet Tracer to connect the Accounts and Delivery departments. The network should allow communication between both departments while using appropriate IP addressing and networking devices.

### Requirements

1. Each department should contain at least two PCs.
2. Appropriate number of routers and switches should be used.
3. The network address 192.168.40.0 should be used for IP addressing.
4. All devices should be connected using appropriate cables.
5. PCs in the Delivery department should be able to communicate with PCs in the Accounts department.

### Network Topology

The network consists of:

- \* One Router (R1)
- \* Two Switches (SW1 and SW2)
- \* Two PCs in the Accounts department
- \* Two PCs in the Delivery department
- \* One printer for each department

## IP Addressing Scheme

### 1. Accounts Department :

Network Address: 192.168.40.0/29

| Device           | IP Address   |
|------------------|--------------|
| -----            | -----        |
| PC1              | 192.168.40.1 |
| PC2              | 192.168.40.2 |
| Printer          | 192.168.40.3 |
| Router Interface | 192.168.40.6 |

Subnet Mask: 255.255.255.248

Default Gateway: 192.168.40.6

### 2. Delivery Department

Network Address: 192.168.40.8/29

| Device           | IP Address    |
|------------------|---------------|
| -----            | -----         |
| PC1              | 192.168.40.9  |
| PC2              | 192.168.40.10 |
| Printer          | 192.168.40.11 |
| Router Interface | 192.168.40.14 |

Subnet Mask: 255.255.255.248

Default Gateway: 192.168.40.14

## Devices Used

1 Cisco Router

2 Cisco Switches

4 PCs

2 Printers

## **Cable Connections**

\* Router to Switch: Copper Straight-Through Cable

\* Switch to PC: Copper Straight-Through Cable

\* Switch to Printer: Copper Straight-Through Cable

## **Configuration Process**

1. Place all network devices in Packet Tracer.
2. Connect devices using appropriate cables.
3. Configure router interfaces with IP addresses.
4. Configure IP addresses for all PCs and printers.
5. Configure default gateways for all end devices.
6. Enable router interfaces using the `no shutdown` command.
7. Verify connectivity between departments using the ping command.

## **Connectivity Testing**

The following tests were performed:

- \* Accounts PC to Accounts Printer.
- \* Delivery PC to Delivery Printer.
- \* Delivery PC to Accounts PC.
- \* Accounts PC to Delivery PC.

All tests were successful, confirming proper routing between the two departmental networks.

## **Result**

A small office network connecting the Accounts and Delivery departments was successfully designed and implemented using Cisco Packet Tracer. The network satisfies all project requirements and provides communication between both departments through a router using two separate /29 subnets.

## **Conclusion**

This project demonstrates the basic concepts of network design, IP addressing, subnetting, router configuration, switch connectivity, and end-to-end communication. The project provides a practical understanding of how different departments in an organization can be connected through a routed network.